

Notes: Innes (JET, 1990)

Limited Liability and Incentive Contracting with Ex-ante Action Choices

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1 Big picture

- **Question.** How does limited liability change the optimal financial contract when an entrepreneur chooses an unobservable *ex-ante* effort level before profit is realized?
- **Setting.** A risk-neutral entrepreneur needs outside funding from a competitive risk-neutral investor. After signing a contract, the entrepreneur privately chooses effort, which shifts the probability distribution of firm profit.
- **Core challenge.** Without liability limits, the classic risk-neutral principal-agent logic gives the investor a fixed payment and leaves the entrepreneur as full residual claimant, thereby inducing the first-best effort level. Limited liability blocks that contract whenever low-profit states can occur.
- **Main conceptual contribution.** The paper shows that liability limits are not a small technical detail. They reshape the contract space and, depending on whether the investor payoff must be monotone in profit, produce two very different optimal security forms.
- **Main result under monotonicity.** If the investor payoff must be weakly increasing in realized profit, the optimal contract is a **standard debt contract**, $B^D(\pi; z) = \min\{\pi, z\}$. The entrepreneur then exerts less than first-best effort.
- **Main result without monotonicity.** If the investor payoff need not be monotone, the optimum becomes a **live-or-die** contract: the investor is paid only in low-profit states and gets nothing in high-profit states. This gives the entrepreneur stronger upside incentives.
- **Efficiency result.** Without the monotonicity constraint, first-best effort may or may not be attainable. It is attainable when a suitably scaled live-or-die contract can both satisfy investor break-even and leave the entrepreneur exactly at the first-best effort choice.
- **Why debt emerges.** Among monotone contracts subject to liability limits, debt concentrates investor claims as much as possible in low-profit states and frees the entrepreneur's residual claim in high-profit states, where extra effort matters most.
- **Information result.** In the monotone setting, the entrepreneur underprovides effort, so information about effort remains valuable. Without monotonicity, first-best can sometimes be restored, in which case the value of monitoring effort can disappear.
- **Broader takeaway.** Debt can be rationalized here *without* costly state verification or adverse selection. It arises because liability limits plus monotonicity make debt the best available way to preserve entrepreneurial effort incentives.

2 Data and measurement (key objects)

2.1 Why this is a theory paper

- This is not an empirical paper. There is no sample, no regression table, and no causal identification design.
- The relevant “objects” are the model primitives and solution objects:

- outside funding need $I > 0$,
- risk-free gross return $1 + p$,
- profit distribution $g(\pi | e)$,
- entrepreneurial effort e ,
- investor payoff function $B(\pi)$,
- first-best effort e^* ,
- debt contracts $B^D(\pi; z)$,
- and live-or-die contracts $B^{LD}(\pi; z)$.

Interpretation. The paper’s “measurement” problem is to define exactly which features of the contract and the profit distribution matter for incentives.

2.2 Technology and profit distribution

Firm profit at date 1 is

$$\pi \in [0, \infty). \quad (1)$$

The profit density and distribution are written as

$$g(\pi | e), \quad G(\pi | e), \quad (2)$$

where e is the entrepreneur’s ex-ante effort choice.

Higher effort improves the profit distribution in the sense of the monotone likelihood ratio property (MLRP):

$$\frac{\partial}{\partial \pi} \left[\frac{g_e(\pi | e)}{g(\pi | e)} \right] > 0, \quad (3)$$

where $g_e(\pi | e) = \partial g(\pi | e) / \partial e$.

The paper also assumes

$$\mathbb{E}[\pi | e = 0] = 0. \quad (4)$$

Interpretation. MLRP says that more effort shifts probability weight toward higher-profit states in a particularly strong way. This is the key statistical condition behind the debt result and the live-or-die result.

2.3 Financing environment

- The entrepreneur must raise a fixed amount $I > 0$ from outside investors.
- Investors are risk-neutral and competitive.
- The alternative safe gross return is $1 + p$.
- The contract specifies an investor payoff schedule $B(\pi)$ contingent on realized profit.

Because investors are competitive, any feasible contract must satisfy an expected return requirement:

$$\mathbb{E}[B(\pi) \mid e] \geq (1 + p)I. \quad (5)$$

Interpretation. The principal here is not an active monopolist who extracts rents. The investor is competitive and just needs to break even at the market return.

2.4 Limited liability and the monotonicity restriction

Limited liability imposes two bounds on the investor payoff:

$$0 \leq B(\pi) \leq \pi \quad \forall \pi. \quad (6)$$

The upper bound means the entrepreneur cannot pay more than the realized profit. The lower bound means the investor cannot be forced to make an additional payment after investing in the firm.

The paper then studies an additional optional restriction:

$$B(\pi + \varepsilon) \geq B(\pi) \quad \forall (\pi, \varepsilon) \in \mathbb{R}_+^2, \quad (7)$$

so the investor payoff is weakly increasing in profit.

The paper gives two motivations for this monotonicity requirement.

- **Sabotage rationale.** If the investor payoff falls in some profit range, investors might want to destroy or divert output to move the firm to a more favorable part of the payoff schedule.
- **Hidden-borrowing rationale.** If the entrepreneur can costlessly inflate the profit report by borrowing right before settlement, any decreasing segment of $B(\pi)$ can be gamed away, so only an equivalent nondecreasing payoff schedule is relevant.

Interpretation. The entire paper turns on whether this monotonicity restriction is imposed. With it, debt wins. Without it, nonmonotone contracts dominate.

2.5 Entrepreneur preferences

Let the entrepreneur's date-1 monetary payoff be

$$w = \pi - B(\pi). \quad (8)$$

Utility is

$$V(w, e) = a(e)w - v(e), \quad (9)$$

with $a(e) > 0$.

Interpretation. Utility is linear in money conditional on effort, so the model abstracts from risk-sharing and isolates incentive effects of liability constraints.

2.6 Effort choice after the contract is signed

Given a contract $B(\pi)$, the entrepreneur chooses effort to solve

$$\max_e \mathbb{E}[V(\pi - B(\pi), e) \mid e] = a(e) \int_0^\infty (\pi - B(\pi))g(\pi \mid e) d\pi - v(e). \quad (10)$$

The paper imposes regularity conditions ensuring existence of an optimal effort choice and, in special contract classes, uniqueness.

Interpretation. The investor cannot contract directly on effort. Contract terms work only through how they change the entrepreneur's marginal gain from shifting probability mass toward high-profit states.

2.7 The entrepreneur's contract-choice problem

The entrepreneur chooses the payoff schedule to maximize expected utility subject to investor break-even, incentive compatibility, and limited liability:

$$\begin{aligned} & \max_{B,e} \mathbb{E}[V(\pi - B(\pi), e) \mid e] \\ \text{s.t. } & \mathbb{E}[B(\pi) \mid e] \geq (1+p)I, \\ & e \text{ solves the entrepreneur's effort problem,} \\ & 0 \leq B(\pi) \leq \pi \quad \forall \pi, \end{aligned} \quad (11)$$

with the monotonicity restriction added when appropriate.

Interpretation. The contract is chosen by the entrepreneur because competitive investors only care about getting their required expected return. Everything else is about preserving effort incentives.

2.8 The first-best benchmark

If limited liability were absent, the entrepreneur could promise the investor a fixed payment $(1+p)I$ in every state and retain full residual claim above that amount. The benchmark effort level therefore solves

$$\max_e \mathbb{E}[V(\pi - (1+p)I, e) \mid e]. \quad (12)$$

The associated first-order condition is

$$a_e(e)(\mathbb{E}[\pi \mid e] - (1+p)I) + a(e)\frac{d\mathbb{E}[\pi \mid e]}{de} - v_e(e) = 0. \quad (13)$$

Denote the unique solution by e^* .

Interpretation. e^* is the right benchmark because it is exactly what a risk-neutral entrepreneur would choose if he could be made the full residual claimant.

3 Models and empirical specifications (all equations + interpretation)

3.1 The entrepreneur's private effort problem

Fix any admissible contract $B(\pi)$. The entrepreneur chooses e to maximize

$$U(e; B) = \mathbb{E}[V(\pi - B(\pi), e) \mid e]. \quad (14)$$

Because higher effort shifts weight toward high-profit states, the slope of $U(e; B)$ depends on how much of those high-profit gains are kept by the entrepreneur versus handed over to the investor.

Interpretation. The contract is not disciplining effort through punishments written directly on e . It is disciplining effort by shaping the entrepreneur's residual claim across profit states.

3.2 The full contracting problem under limited liability

The entrepreneur's optimization program is

$$\begin{aligned} \max_{B, e} \quad & a(e) \int_0^\infty (\pi - B(\pi)) g(\pi \mid e) d\pi - v(e) \\ \text{s.t.} \quad & \int_0^\infty B(\pi) g(\pi \mid e) d\pi \geq (1 + p)I, \\ & e \in \arg \max_{\tilde{e}} \mathbb{E}[V(\pi - B(\pi), \tilde{e}) \mid \tilde{e}], \\ & 0 \leq B(\pi) \leq \pi \quad \forall \pi, \end{aligned} \quad (15)$$

plus $B(\pi + \varepsilon) \geq B(\pi)$ when the monotonicity restriction is imposed.

Interpretation. This is a principal-agent problem written in a finance language: a security design problem with hidden action and statutory liability limits.

3.3 Debt contracts under the monotonicity restriction

Define a standard debt contract by

$$B^D(\pi; z) = \min\{\pi, z\}, \quad z > 0. \quad (16)$$

Suppose some monotone nondebt contract $B^{\text{ND}}(\pi)$ with induced effort e^{ND} were optimal. Let

$$Q \equiv \mathbb{E}[B^{\text{ND}}(\pi) \mid e^{\text{ND}}], \quad (17)$$

and choose z_0 so that

$$\mathbb{E}[B^D(\pi; z_0) \mid e^{\text{ND}}] = Q. \quad (18)$$

This comparison debt contract gives investors the same expected return at the nondebt effort level, but it pays investors *more* in low-profit states and *less* in high-profit states than the nondebt monotone payoff.

Interpretation. This comparison isolates a pure incentive effect. The expected transfer to investors is held fixed at e^{ND} , but the state pattern of payments is changed.

3.4 Why shifting payments downward strengthens effort

A central technical lemma shows: if one contract shifts investor payments toward low-profit states and away from high-profit states, then under MLRP investors capture a smaller fraction of the marginal return to effort.

The paper's key derivative for debt is

$$\frac{\partial}{\partial e} \mathbb{E}[B^D(\pi; z) \mid e] = - \int_0^z G_e(\pi \mid e) d\pi > 0. \quad (19)$$

So investor expected payoff under debt rises when effort rises.

The key comparison result is then

$$e^D(z_0) > e^{ND}. \quad (20)$$

Interpretation. Debt works because it lets the entrepreneur keep more of the marginal payoff in high-profit states. Since high-profit states become more likely when effort rises, debt strengthens incentives relative to any other monotone contract that gives investors more upside participation.

3.5 Proposition 1: the optimal monotone contract is debt

Under an existence condition ensuring that some debt contract can satisfy investor break-even, the solution to the monotone problem has three properties:

1. $B(\pi) = B^D(\pi; z) = \min\{\pi, z\}$ for some $z > 0$;
2. investor break-even binds, so $\mathbb{E}[B(\pi) \mid e] = (1 + p)I$;
3. the induced effort is strictly below first-best, $e < e^*$.

Interpretation. Debt is the best monotone way to commit the entrepreneur to work hard, but monotonicity still forces some sharing of marginal effort benefits with investors, so effort remains inefficiently low.

3.6 Economic intuition for the debt result

The paper's logic can be compressed into one sentence: *among monotone contracts, debt puts as much of the investor claim as possible into the states where effort matters least for the entrepreneur's private payoff.*

Why?

- If a monotone contract keeps rising in high-profit states, the investor shares in the upside from extra effort.
- The entrepreneur still bears the full disutility of effort.
- So any monotone contract with too much investor upside softens incentives.
- Debt caps the investor claim and leaves all additional upside to the entrepreneur once profit is above the debt face value.

Interpretation. This paper gives a clean incentive-based reason that debt is common even in a very general contract space.

3.7 The nonmonotone benchmark: live-or-die contracts

Without the monotonicity restriction, the paper defines the extreme contract

$$B^{\text{LD}}(\pi; z) = \begin{cases} \pi, & \pi \leq z, \\ 0, & \pi > z. \end{cases} \quad (21)$$

The investor is paid only in low-profit states and receives nothing in high-profit states.

Interpretation. This is the polar opposite of equity-like upside sharing. It maximizes the entrepreneur's claim on the high-profit states that are especially sensitive to effort.

3.8 The relaxed first-order-condition problem

To characterize the unrestricted optimum, the paper studies Rogerson's relaxed first-order-condition problem:

$$\begin{aligned} & \max_{B,e} \mathbb{E}[V(\pi - B(\pi), e) \mid e] \\ & \text{s.t. } \mathbb{E}[B(\pi) \mid e] \geq (1+p)I, \\ & \quad \frac{d}{de} \mathbb{E}[V(\pi - B(\pi), e) \mid e] \geq 0, \\ & \quad 0 \leq B(\pi) \leq \pi \quad \forall \pi. \end{aligned} \quad (22)$$

If a solution to this relaxed problem also satisfies the entrepreneur's true incentive constraint, then it solves the original problem.

Interpretation. The relaxed problem is a shortcut that turns the hidden-action constraint into a tractable first-order condition.

3.9 Proposition 2: if incentives still bind, pure live-or-die is optimal

Let μ denote the multiplier on the relaxed incentive constraint. If $\mu > 0$, so that incentives bind in the relaxed problem, then the optimal solution has the following properties:

1. $B(\pi) = B^{\text{LD}}(\pi; z)$ for some $z > 0$;
2. investor break-even binds;
3. effort is below first-best, $e < e^*$;
4. and, under the paper's regularity assumption, this solution also solves the original contracting problem.

Interpretation. Once monotonicity is dropped, the optimizer becomes bang-bang: maximum investor extraction in low states, zero extraction in high states.

3.10 When first-best can be restored without monotonicity

Now suppose the relaxed incentive constraint is slack, $\mu = 0$. Then the paper shows two key facts:

1. effort must equal the first-best level, $e = e^*$;
2. investor break-even still binds.

The paper next proves that there is a unique cutoff $z^* > 0$ such that

$$\left. \frac{\partial}{\partial e} \mathbb{E}[B^{\text{LD}}(\pi; z^*) \mid e] \right|_{e=e^*} = 0. \quad (23)$$

If there exists a scaling factor $\alpha^* \in (0, 1]$ satisfying

$$\alpha^* \mathbb{E}[B^{\text{LD}}(\pi; z^*) \mid e^*] = (1 + p)I, \quad (24)$$

then the scaled live-or-die contract

$$B(\pi) = \alpha^* B^{\text{LD}}(\pi; z^*) \quad (25)$$

supports first-best effort and satisfies investor break-even.

Interpretation. First-best is possible when the outside funding need is small enough relative to what a suitably calibrated low-state claim can raise at the first-best effort level.

3.11 Proposition 4 and the final classification

Putting the pieces together, the unrestricted problem has two possible optimal forms.

- **Case 1: first-best feasible.** If there exists $\alpha^* \in (0, 1]$ satisfying the condition above, the optimum is a *scaled live-or-die* contract and effort equals e^* .
- **Case 2: first-best infeasible.** If no such α^* exists, the optimum is a *pure live-or-die* contract and effort remains below e^* , though it is still higher than under the monotone debt solution.

The paper's final corollary is that the unrestricted optimum is necessarily nonmonotone.

Interpretation. Once the monotonicity restriction is removed, the entrepreneur always wants a payoff structure that violates it. That is the deepest contrast with the debt result.

3.12 Value of information and comparative statics

The paper emphasizes one more implication.

- In the monotone debt case, effort is strictly suboptimal, so direct information about effort would be valuable.
- In the unrestricted case, first-best may obtain, so information about effort may have zero value.

The paper also notes the comparative-static logic behind the first-best feasibility condition:

- a lower outside funding requirement I makes first-best easier to sustain;

- a larger expected low-state payment under the calibrated live-or-die contract also makes first-best more likely.

Interpretation. Liability constraints do not mechanically destroy efficiency. Whether they do so depends on how much financing has to be raised and how flexibly the contract can allocate claims across states.

4 Identification and instruments (what makes the theoretical results credible)

4.1 No empirical identification or instruments

- The paper does not identify a treatment effect from data.
- There is no instrument, difference-in-differences design, regression discontinuity, or structural estimation exercise.
- Its credibility comes from the transparency of the assumptions and from the sharp contrast between the monotone and nonmonotone contract spaces.

Interpretation. The relevant question here is not “what identifies a coefficient?” but “which assumptions generate the contract form?”

4.2 The key assumptions that do the work

1. **Risk neutrality.** Both sides are risk-neutral, so the paper can isolate incentive effects and ignore risk-sharing motives.
2. **Competitive investors.** The investor’s problem is reduced to expected break-even at the safe rate.
3. **MLRP.** Higher effort must disproportionately increase the likelihood of high-profit states.
4. **Observable ex-post profit.** The contract can depend on realized profit, even though effort is hidden.
5. **Limited liability.** Payoffs must satisfy $0 \leq B(\pi) \leq \pi$.
6. **Optional monotonicity.** When imposed, it rules out exactly the nonmonotone payoff shapes that would otherwise sharpen incentives.

Interpretation. The debt result is not just a generic consequence of hidden action. It is the interaction of hidden action, liability limits, and monotonicity.

4.3 Why MLRP is central

The entire comparison between debt and nondebt contracts relies on MLRP.

- MLRP ensures that higher effort pushes relatively more probability weight toward higher-profit outcomes.
- Therefore, shifting investor claims out of high-profit states and into low-profit states always strengthens the entrepreneur’s private incentive to work.
- Without MLRP, the ranking of contract forms by effort incentives would not be so clean.

Interpretation. MLRP is the engine under the hood. It is what turns “more high-state upside” into “more effort.”

4.4 Why the monotonicity restriction changes everything

The monotonicity restriction is not a decorative assumption.

- It forces investor payoffs to rise with profit.
- That means the investor must share in at least some of the upside generated by effort.
- Once this restriction is lifted, the entrepreneur wants the exact opposite shape: investor claims concentrated in low states and removed from high states.

Interpretation. Debt is optimal only within the monotone contract class. It is not optimal in the unrestricted contract class.

4.5 What the model does and does not explain

The paper explains why debt can arise from incentive considerations under limited liability, but it abstracts from several important issues.

- There is no risk aversion, so the analysis does not balance risk sharing against incentive provision.
- There is no asymmetric information about project quality or entrepreneur type in the base model.
- Investment scale and investment policy are fixed rather than chosen jointly with the contract.
- The paper does not model costly verification or bankruptcy courts, so its debt result is distinct from Townsend/Gale-Hellwig style arguments.

Interpretation. The paper is best read as a benchmark mechanism. It shows one clean route from liability limits to debt without claiming that this is the only route.

4.6 Limits discussed by the author

The conclusion highlights four concrete limitations.

1. Entrepreneur and investor are both risk-neutral.
2. Beliefs about the profit distribution are symmetric.
3. Investors know the entrepreneur’s preferences and effort-response mapping.

4. Investment level and investment policy are treated as fixed and known.

The paper also notes an important extension: if the entrepreneur can commit to place some borrowed funds in a riskless bond, the monotone debt result survives, but in the nonmonotone environment first-best can always be achieved because the lower-bound liability constraint is effectively relaxed.

Interpretation. The upper-bound liability limit $B(\pi) \leq \pi$ is enough to generate the debt result under monotonicity, but the lower-bound limit $B(\pi) \geq 0$ matters for whether nonmonotone contracts can still leave effort inefficient.

4.7 Relation to the finance literature

The paper positions itself against other theories of capital structure.

- Relative to standard principal-agent models without liability limits, the paper explains why the full-residual-claimant contract becomes infeasible.
- Relative to costly-state-verification models, debt here does not arise from verification costs.
- Relative to adverse-selection models, debt here does not arise from signaling or sorting.
- Relative to Jensen-Meckling style intuition, the paper formalizes the idea that financing claims shape managerial incentives.

Interpretation. The contribution is conceptual: debt can emerge from limited liability and incentive preservation alone.

5 Tables and figures

This paper has only one conventional figure and no empirical tables. The most useful way to read the evidence is therefore: the actual figure from the paper plus summary exhibits that collect the theoretical results.

5.1 Figure 1: debt versus a generic monotone nondebt contract

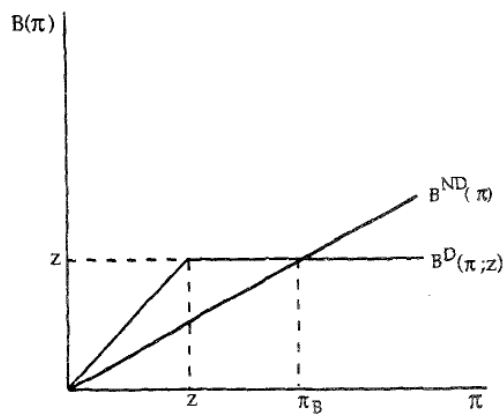


FIGURE 1

Figure 1: Debt versus a monotone nondebt contract

Read. The figure compares a debt payoff $B^D(\pi; z)$ with a monotone nondebt payoff $B^{ND}(\pi)$. Relative to the nondebt contract, debt pays the investor more in low-profit states and less in high-profit states.

Economic message. This picture is the geometric core of the debt result. Under MLRP, moving investor claims downward like this strengthens entrepreneurial effort incentives, which is why debt dominates other monotone contracts.

5.2 Exhibit 1: Core model objects

Table 1: Main objects in the model

Object	Definition	Economic meaning
I	External funds required	How much outside finance the entrepreneur must raise.
p	Safe net rate of return	Investor opportunity cost of capital.
$B(\pi)$	Investor payoff schedule	The financial contract.
$g(\pi e)$	Profit density conditional on effort	How effort changes firm performance.
e	Entrepreneur effort	Hidden ex-ante action.
e^*	First-best effort	Benchmark effort with full residual claimancy.
$B^D(\pi; z)$	$\min\{\pi, z\}$	Standard debt contract.
$B^{LD}(\pi; z)$	π if $\pi \leq z$, else 0	Live-or-die contract.

Read. The model has very few primitives. Most of the economics comes from how liability limits restrict $B(\pi)$.

Economic message. This is a paper about the shape of the payoff schedule, not about heterogeneity in investors, balance-sheet frictions, or market power.

5.3 Exhibit 2: Optimal contract by environment

Table 2: Optimal contracts across environments

Environment	Feasible class	Optimal contract	Effort outcome	Main intuition
No liability limits	All payoff schedules, including fixed investor payment in every state	Fixed payment to investor; entrepreneur is full residual claimant	$e = e^*$	Full marginal return to effort stays with entrepreneur.
Limited liability + monotonicity	$0 \leq B(\pi) \leq \pi$ and $B'(\pi) \geq 0$ in the weak sense	Debt $B^D(\pi; z)$	$e < e^*$	Debt minimizes investor upside participation within the monotone class.
Limited liability, no monotonicity, first-best feasible	Only $0 \leq B(\pi) \leq \pi$	Scaled live-or-die $\alpha^* B^{LD}(\pi; z^*)$	$e = e^*$	Enough low-state claims can be raised without touching high-state upside.
Limited liability, no monotonicity, first-best infeasible	Only $0 \leq B(\pi) \leq \pi$	Pure live-or-die $B^{LD}(\pi; z)$	$e < e^*$	Pure low-state extraction gets the entrepreneur as close to first-best as possible.

Read. There are really four economic environments, and only one of them delivers ordinary debt.

Economic message. Debt is not “the” universal optimum; it is the optimum only when liability limits are combined with the monotonicity restriction.

5.4 Exhibit 3: Proposition map

Table 3: What each main result establishes

Result	Formal content	Economic meaning
Proposition 1	Under monotonicity, the optimum is debt and effort is below first-best.	Debt is the best incentive-preserving contract inside the monotone class, but liability limits still create underprovision of effort.
Proposition 2	In the unrestricted relaxed problem, if incentives bind then the optimum is pure live-or-die.	Once nonmonotone payoffs are allowed, upside claims move entirely off the investor's payoff schedule.
Proposition 3	If a scaled live-or-die contract can meet break-even at the calibrated cutoff z^* , it supports first-best.	Nonmonotone contracts can sometimes fully undo the inefficiency created by liability limits.
Proposition 4	The multiplier on the relaxed incentive constraint is zero iff the first-best-supporting scaled live-or-die contract exists.	First-best feasibility is exactly characterized by whether low-state extraction can finance the investor at e^* .

Read. The propositions form a clean decision tree: monotone gives debt; unrestricted gives live-or-die, sometimes scaled enough to recover first-best.

Economic message. The paper's conclusions are sharp and exhaustive once the contract class is fixed.

5.5 Exhibit 4: Efficiency and information

Table 4: Effort efficiency and the value of monitoring

Case	Effort relative to first-best	Value of information about effort
Monotone debt optimum	Strictly below first-best	Positive, because monitoring could improve incentives or permit a better contract.
Unrestricted optimum with first-best feasible	Exactly first-best	Potentially zero, because incentives are already fully aligned.
Unrestricted optimum with first-best infeasible	Below first-best, but stronger than under monotone debt	Positive, though less obviously than in the debt case because the contract is already highly incentive intensive.

Read. The paper connects contract form to the value of information.

Economic message. Better monitoring is not intrinsically valuable; it is valuable only because feasible contracts leave effort distorted.

6 Short summary

- The paper studies a hidden-action financing problem with risk-neutral parties and statutory limited liability.
- Without liability limits, a fixed investor payment would induce the first-best effort level.
- With liability limits and a monotone payoff restriction, the optimal contract is standard debt and effort is inefficiently low.
- Without the monotonicity restriction, the optimal contract is nonmonotone: a live-or-die payoff that taxes only low-profit states.
- If a suitably scaled live-or-die contract can satisfy investor break-even at the first-best effort level, efficiency is restored; otherwise the optimum is still live-or-die but second-best.
- The paper therefore gives a clean theory of why debt arises in one contract class and why it is not globally optimal in a larger contract class.

7 Conclusion

7.1 Core conclusion

The paper's central conclusion is that limited liability changes the optimal incentive contract in a fundamental way. When investor payoffs must be monotone in realized profit, the optimal security is standard debt and it delivers less than first-best effort. When the monotonicity restriction is removed, the contract that best preserves effort incentives becomes nonmonotone and takes a live-or-die form.

7.2 Economic intuition

Effort is valuable because it shifts probability weight toward high-profit states. The best contract for incentives is therefore the one that leaves the entrepreneur with as much high-state upside as possible while still raising the required outside finance. Debt is the best way to do that inside the monotone class; live-or-die contracts do even better once monotonicity is abandoned.

7.3 Takeaway

This paper is a foundational reminder that contract form cannot be separated from legal and technological constraints. Debt is optimal here not because it is assumed, but because limited liability plus monotonicity make it the least damaging way to finance the investor while preserving entrepreneurial incentives.